

JEE MAIN REVISION MASTER: SETS AND RELATIONS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 117 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Equivalence Relations (Reflexive, Symmetric, Transitive)	78 PYQs	Relations
Very High	Set Operations & Venn Diagram Principle	44 PYQs	Set Theory

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Equivalence Relations (Reflexive, Symmetric, Transitive)	$\text{Reflexive: } \forall a \in A, (a,a) \in R, \quad \text{Symmetric: } (a,b) \in R \implies (b,a) \in R, \quad \text{Transitive: } (a,b), (b,c) \in R \implies (a,c) \in R$
Set Operations & Venn Diagram Principle	$n(A \cup B) = n(A) + n(B) - n(A \cap B), \quad n(A \cup B \cup C) = \sum n(A) - \sum n(A \cap B) + n(A \cap B \cap C)$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Equivalence Relations (Reflexive, Symmetric, Transitive) (Very High Yield — 78 PYQs)

Core Principle: Formal definitions of binary relations on sets, properties for equivalence, and total number of reflexive relations $2^{n^2} - n$.

■ **Caution / JEE Pitfall:** A relation is vacuously transitive if there are no pairs (a,b) and (b,c) with distinct elements to contradict transitivity.

■ **Speed Trick:** Total number of relations on set of n elements is 2^{n^2} ; total number of symmetric relations is $2^{n(n+1)/2}$.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Consider the relation R on the set $-2, -1, 0, 1, 2$ defined by $(a,b) \in R$ if and only if $1+ab > 0$. Then, among the statements : I. The ..." [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "The number of relations, defined on the set a,b,c,d , which are both reflexive and symmetric, is equal to:..." [Ans: N/A]
- [2026 Morning Shift] Q#2 (MCQ): "Let R be a relation defined on the set $1,2,3,4 \times 1,2,3,4$ by $R = \{(a,b), (c,d) : 2a+3b=3c+4d\}$. Then the number of elements in R is..." [Ans: N/A]

3.2 Set Operations & Venn Diagram Principle (Very High Yield — 44 PYQs)

Core Principle: Inclusion-exclusion principle for cardinality, De Morgan's set identities, and power set cardinality 2^n .

■ **Caution / JEE Pitfall:** If set A has n elements, the number of proper subsets is $2^n - 1$ (excluding the set itself).

■ **Speed Trick:** Number of elements belonging to EXACTLY two of three sets: $\sum n(A \setminus B) - 3n(A \cap B \cap C)$.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "The number of elements in the relation $R = \{(x, y) : 4x^2 + y^2 < 52, x, y \in \mathbb{Z} \text{ is...} \}$ [Ans: N/A]
- [2026 Evening Shift] Q#1 (Numerical): "Let S be the set of the first 11 natural numbers. Then the number of elements in $A = B \subseteq S : n(B) \equiv 2$ and the product of all elements of B is even is ____." [Ans: 1979]
- [2026 Evening Shift] Q#3 (MCQ): "Let $A = \{x : |x^2 - 10| \leq 6\}$ and $B = \{x : |x - 2| > 1\}$. Then..." [Ans: B]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 22nd January Evening Shift (Q#1)

Let S be the set of the first 11 natural numbers. Then the number of elements in $A = B \subseteq S : n(B) \equiv 2$ and the product of all elements of B is even is ____.

Correct Answer: 1979

Solution Analysis: Let $S = \{1, 2, \dots, 11\}$. The product of all elements of B is even iff B contains at least one even number. Evens in S : 5 (2, 4, 6, 8, 10) Odds in S : 6 Number of subsets containing at least one even: $2^{11} - 2^6 = 2048 - 64 = 1984$ (we subtract the subsets made only of odds). Now enforce $n(B) \geq 2$: subtract the 5 singleton subsets 2, 4, 6, 8, 10. $1984 - 5 = 1979$

Benchmark Problem 2: JEE Main 2025 (Online) 22nd January Evening Shift (Q#1)

Let $A = \{1, 2, 3\}$. The number of relations on A, containing (1, 2) and (2, 3), which are reflexive and transitive but not symmetric, is ____.

Correct Answer: 3

Solution Analysis: Transitivity $(1, 2) \in R, (2, 3) \in R \Rightarrow (1, 3) \in R$ For reflexive $(1, 1), (2, 2), (3, 3) \in R$ Now $(2, 1), (3, 2), (3, 1)$ cannot be taken (1) (2, 1) taken and (3, 2) not taken (2) (3, 2) taken and (2, 1) not taken (3) Both not taken therefore 3 relations are possible.